The Role of the Rhapsody Integration Engine

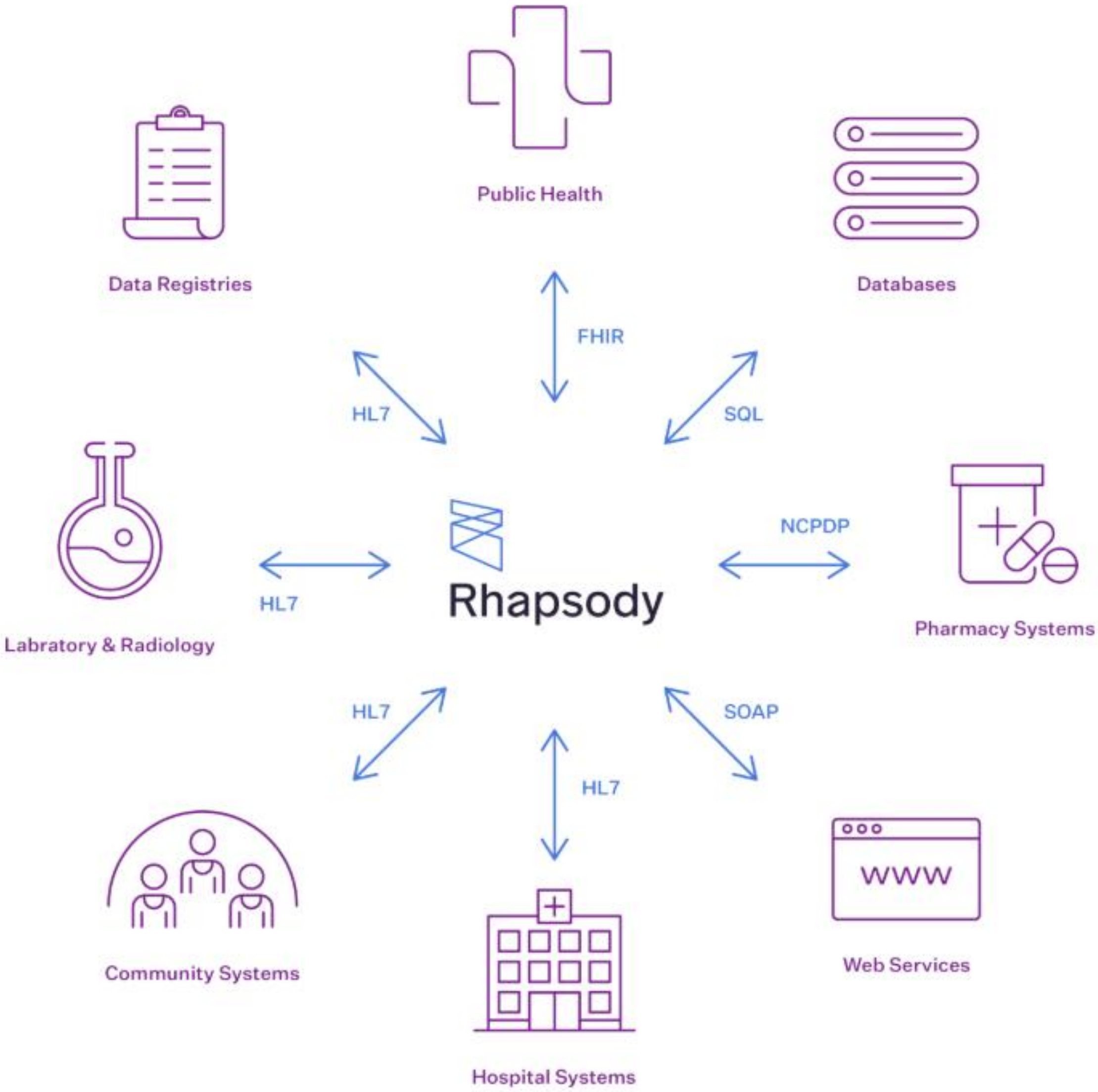
A typical healthcare organization is comprised of many systems providing a wide variety of services. These systems need to communicate with each other to exchange information. For example:

- A patient demographics system that updates other systems with details of currently admitted patients.
- A laboratory system that publishes results for inclusion in the patient record and decision support systems.

Many of these systems use incompatible messaging protocols or cannot connect to all required recipients of the messages they produce. In addition, as healthcare organizations evolve, new systems are added. The ability to manage communications to these systems from a single location makes management of messaging within the organization simpler.

The Rhapsody Integration Engine is designed to connect to many different systems using a wide range of protocols. The messages received by the engine can be manipulated to convert them between the various messaging standards used by each connected system.

Support for a variety of health sector communication and messaging standards is included in the engine, providing solutions to the “many-to-many” problems encountered in Healthcare IT.



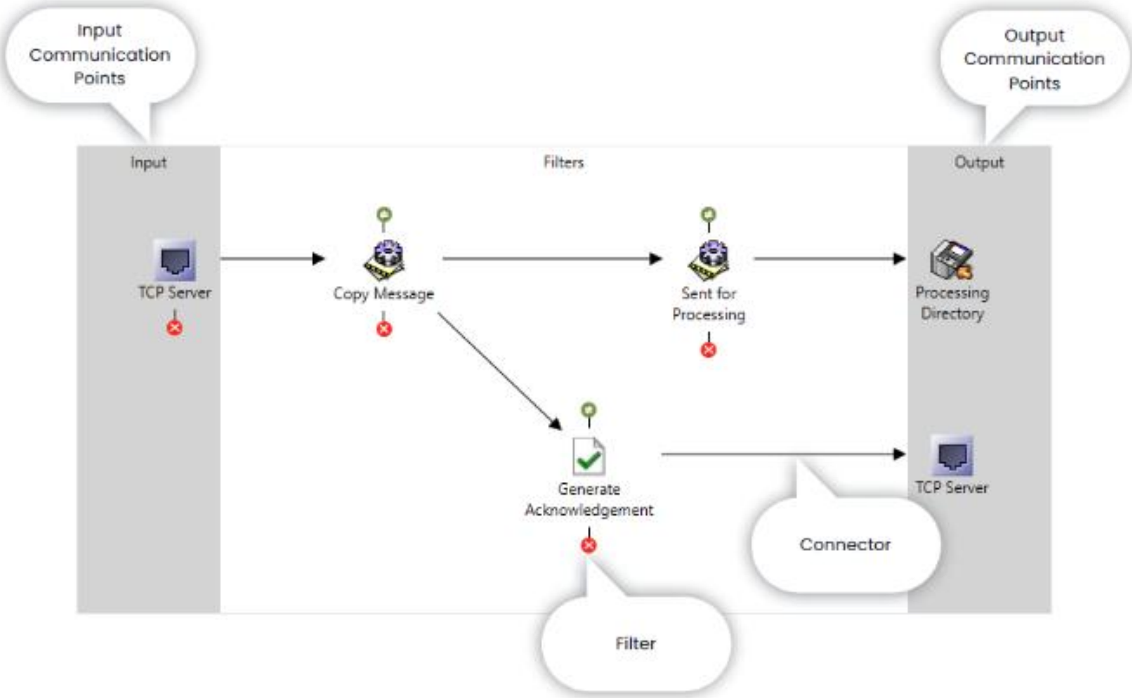
Multi-point Integration Tool

The Rhapsody engine provides a multi-point integration tool within a network of varying systems, as shown in the illustration above. This creates an environment that is more easily managed than those relying on single point-to-point solutions.

Configuration Concepts

At its core, Rhapsody handles messages (blocks of data) that it receives via connections with remote systems. These messages are passed through processing pipelines where business logic can be applied, and ultimately transmitted to remote systems.

Within the engine, four components are used in a basic configuration to process messages.



Rhapsody components

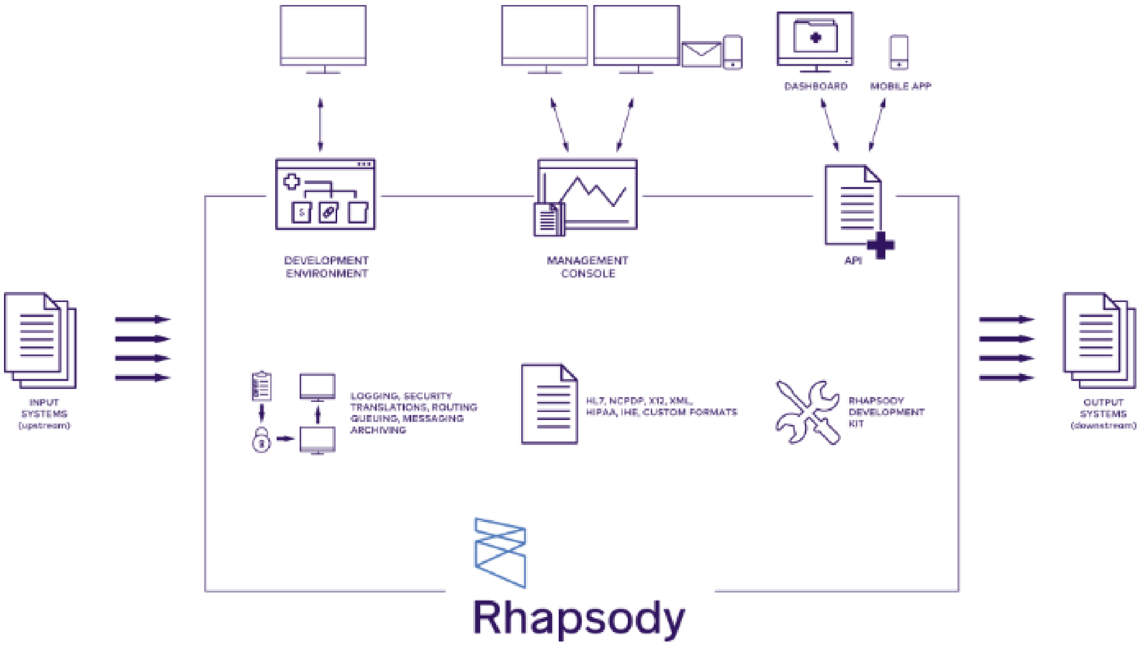
- Communication Points handle connections with remote systems, supporting many different protocols.
  - Communication points that appear on the input side of a configuration receive messages.
  - Communication points on the output side send messages.
  - A communication point that appears on both the input and output sides of a configuration will both send and receive messages.
- Routes are pipelines through the engine traversed by messages. As the message moves through one or more routes, business logic is applied to the message and the path may change depending on the content of the message.
- Filters are sub-components of routes which apply business logic to a message. A route may contain several filters, each of which performs an action on the message (up to and including completely changing the message content and structure). After the processing in a filter is complete, the message path through the route is controlled by the filter connections and connection logic.
- Connectors define the path between components in the route and optionally may establish conditions to be met before the message can traverse the connection.

The Rhapsody Installation

Installing Rhapsody is a two-part process; one for the Rhapsody Engine, which runs in the background, and the other for the Rhapsody User Interface components:

- Rhapsody Integrated Development Environment (IDE)
- Rhapsody Messaging Toolkit
- Management Console

We will discuss each component in more detail later in this module.



Rhapsody components

Rhapsody Documentation

The Rhapsody IDE, Messaging Toolkit and Management Console are supplied with a full set of administration and installation manuals.

Select the **Help** button on any Rhapsody IDE dialog or in the Management Console to open the content in the Rhapsody Documentation Portal. Here you can view up-to-date, context-sensitive information. If you are using Rhapsody in an environment where there is no internet access, offline help can be installed and configured.

Rhapsody Documentation Portal

View the documentation portal.

[Open docs]

Offline Help

How to install and configure offline help.

[Open docs]

Navigating the documentation

When documentation is opened from within Rhapsody, content relating to the dialog or screen you are currently viewing is displayed.

Intended Use

The Rhapsody Integration Engine has been designed to solve a particular range of problems. These are described in the intended use statement.

Key points for intended use

1. 1

Rhapsody is an external system similar to a Database or Web Service call

As in most external systems, there is no guarantee that a call will succeed. Call failure reasons include the non-availability of dependent resources such as databases or the loss of access to remote resources due to network issues. Applications should be developed with this in mind to ensure they are resilient in the face of failures and that appropriate actions are available dependent on requirements:

- If a request is intended to provide critical information to the application, a suitable error should be displayed to the user and any associated operations should fail.
- If a request is intended to provide supporting information, then suitable warnings should be displayed stipulating that the request did not succeed and that subsequent operations may be adversely affected.

In each case, suitable timeouts should be employed and in line with the responsiveness requirements of the operation/application.

A common example results from using Rhapsody in a clinical workflow situation. This is usually not a problem provided that the above considerations are kept in mind and that the outcome of the workflow does not result in real-time decision making with an immediate impact on patient safety. In clinical workflow scenarios where decisions do not have a real-time requirement, applications that make use of Rhapsody should make use of the above error handling.

2. 2

As Rhapsody requests are asynchronous in nature, there are no expectations that a request will result in an immediate response

Rhapsody requests are asynchronous, therefore responses may not be timely in nature. Rhapsody use should be questioned if a response time of a few seconds is required, taking into account any error handling for failed responses. Again, this may occur in clinical workflow situations but is more likely in an automated workflow containing a time-critical element.

3. 3

Critical systems should never be reliant on one system and should always include human intervention

There may be situations where Rhapsody is used for notification events of a critical nature that demand an immediate response. This is a valid intended use ONLY if the notification events in Rhapsody are not the primary system of alert. This solution is additive to any existing form of alerts that are currently employed and alerts of a critical nature should always have some form of human intervention. However, it is acceptable for Rhapsody to act as a secondary form of notification which augments an existing solution.

4. 4

Rhapsody is not intended to be used to interface directly with medical devices for the purposes of controlling those devices or using information from them for real-time decision making

It is not intended use for Rhapsody to be used to directly control medical devices or to gather data from medical devices which then leads to some form of automated and immediate response that triggers some action.

It is acceptable for Rhapsody to gather data from medical devices which is then stored and used in clinical workflows for later processing, provided that gathering this data does not have immediate ramifications relating to patient safety. Gathering data for analytic purposes is an acceptable use case.

The Rhapsody Engine

The Rhapsody Engine is a platform-independent, high performance Java application that runs as a background process, (either a service(opens in a new tab) on Windows, or a daemon(opens in a new tab) on Unix-like systems). The engine is typically configured with its own disk environment, such as a SAN (Storage Area Network).

Because the engine has no user interface, all configuration and management is performed using external components. The engine implements the following:

- Communications

- Mapping
- Message delivery
- Data transformation and routing of data

As discussed in the previous lesson, messages enter the engine via input communication points, and then pass through a series of filters before exiting the engine via output communication points. The engine analyzes the incoming messages to determine which steps, procedures, or filters need to be applied in handling the message and sending it to its destination. A filesystem-based datastore stores the engine configuration and state information, as well as the message archive.

Rhapsody Service Monitor

When the engine is installed on a Windows machine, the Rhapsody Service Monitor is added to the system tray on the right side of the taskbar. It is used to:

- Quickly identify the status of the engine
- Open the Management Console
- Control the start state of the engine

The Service Monitor icon showing the various engine statuses:

Right-click on the **Service Monitor** icon in the system tray to display the following options, depending on the status of the service. If the icon is not visible, run *\Program Files\Rhapsody\Rhapsody Engine 6\bin\monitor-start.bat* on the Rhapsody server to view it.

Option	Description
Start Rhapsody Service	Starts the Rhapsody service.
Stop Rhapsody Service	Stops the Rhapsody service and closes the Rhapsody IDE.
Open Management Console	Opens the login screen of the Rhapsody Management Console, where you can view the status and activity of all components. This is the default action if you double-click the **Service Monitor** icon.
Close Rhapsody Service Monitor	Stops displaying the Rhapsody Service Monitor.

Security

Groups

Users can be provided with different levels of access. For example, a developer might have full access to the Rhapsody IDE, while a help desk user might have read-only access to the Management Console. This access is based on the groups each user belongs to.

- A group defines the set of actions a user can perform (for example, connecting to the Rhapsody IDE, or viewing the body of a message in the Management Console).
- Group permissions apply to all components within Rhapsody or a locker (more on lockers below).
- Mulitiple groups can be configured in Rhapsody depending on your organization's needs, and a user can be assigned to one or more groups.

After logging in, the functionality a user can access depends on the groups they were assigned to; this is controlled within the IDE's User Manager or via an external LDAP user repository, such as Active Directory.

Lockers

Lockers are another security mechanism to control user access.

- Lockers group related configuration items together within the engine to manage access.
- Security can be applied to a locker to control how certain groups of users interact with the locker. For example, preventing members of the support team from viewing the contents of messages in a mental health reporting locker.
- In most environments, very few lockers are needed as there is no need to restrict access to parts of the configuration or to the messages being processed by that configuration.

We discuss Lockers in more detail in the "Working with IDE" module.

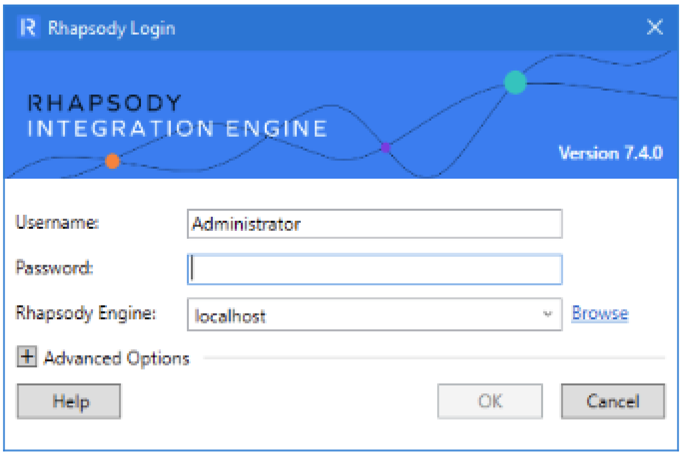
The Rhapsody IDE

The Rhapsody IDE provides the tools and options to configure a Rhapsody instance. It is a Windows client application installed on the PC of each Rhapsody administrator.

To access the Rhapsody IDE

The Rhapsody IDE can be launched from the Windows Start menu (search for Rhapsody IDE).

Many organizations maintain multiple installations of the engine for different purposes, for example, Development, Training and Production. When starting the IDE, you can choose which engine to connect to.



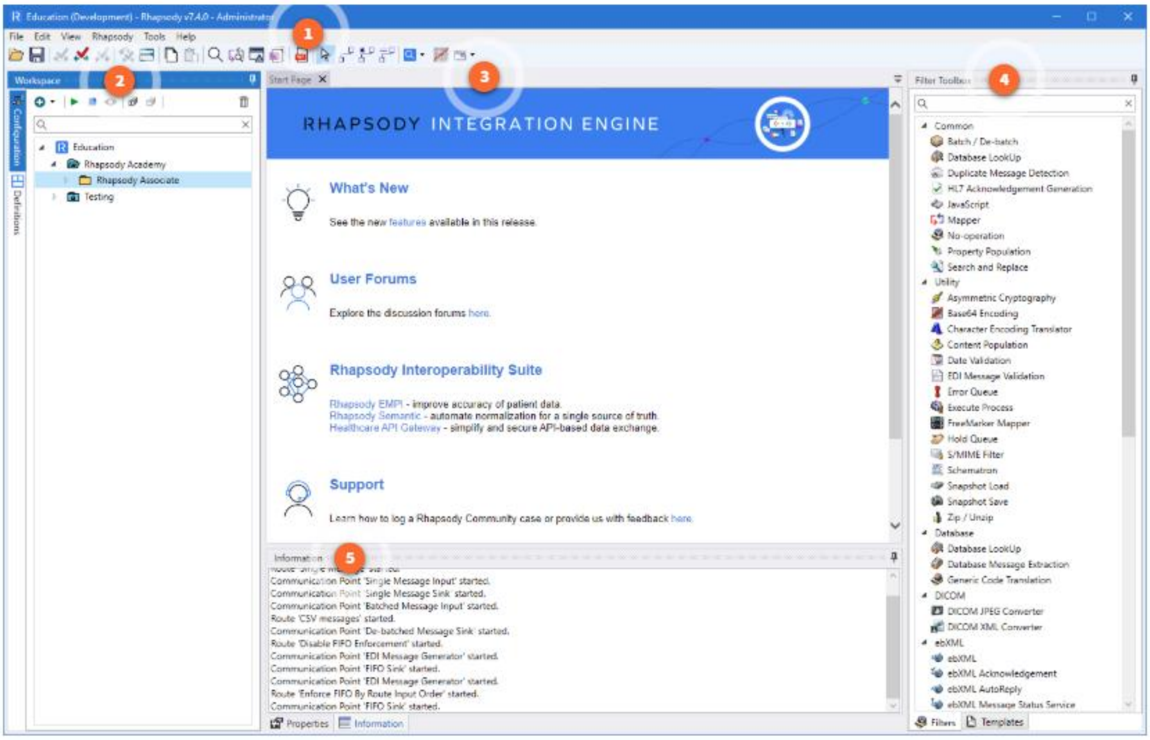
Rhapsody IDE login screen

Rhapsody IDE login screen

After you log in, the functionality you can access depends on the security groups to which you have been assigned. If the connection between the IDE and engine is lost, or the engine is restarted, the IDE will exit and any changes that have not been checked into the engine will auto-save locally.

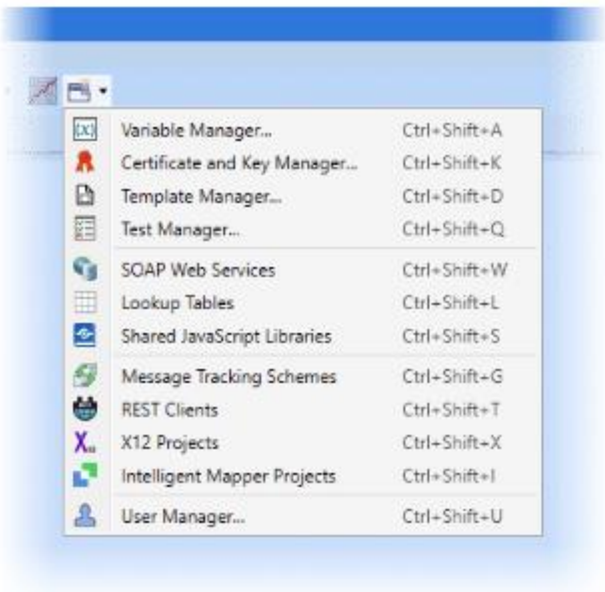
General Layout

When you first log in to the IDE, you will see the Start Page containing several panels described in detail below.



The **Managers** drop-down on the IDE toolbar lists commonly used Rhapsody tools.

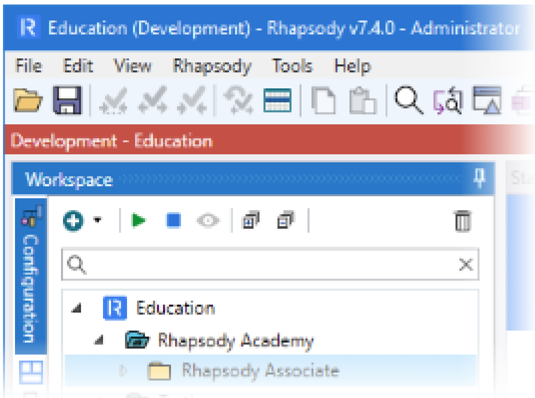
During the Associate course we will make use of several of these tools. The full list of tools, including those that do not appear in the **Managers** drop-down, can be found under the **View** menu.



Managers drop-down

You can configure an [environment indicator(opens in a new tab)] below the toolbar to help identify the engine you are connected to, for example, **Development - Education**.

The background color of the environment indicator can be customized on a per-engine basis.

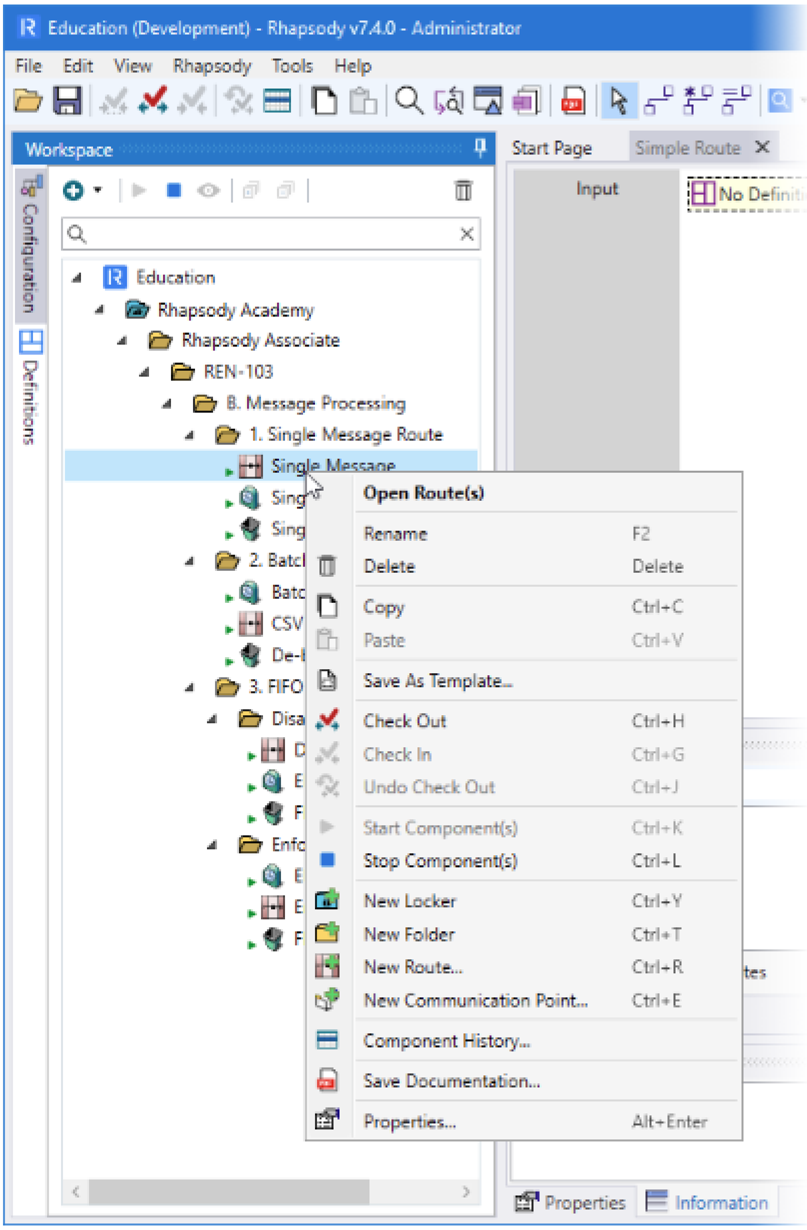


A red environment indicator

The Workspace

The IDE **Workspace** displays all the components in your Rhapsody configuration. The two tabs on the left of the **Workspace** let you view different aspects of your configuration. Use the search box to filter components by name.

- The **Configuration** tab lists the routes and communication points.
- The **Definitions** tab lists all message and mapper definitions in the configuration.



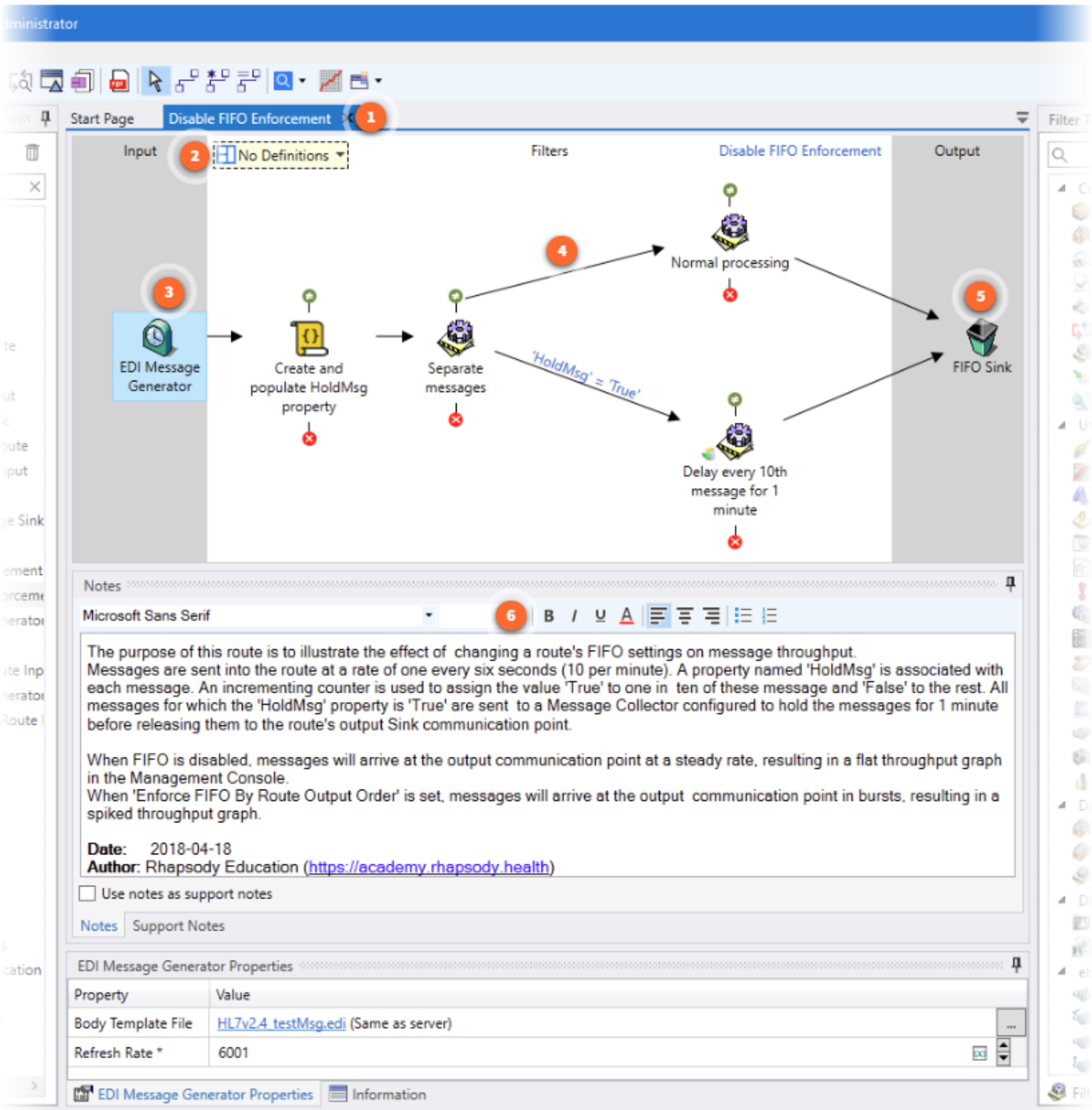
The IDE Workspace showing the right-click menu for the single message route.

Select a component (or multi-select several components) and use the icons at the top of the **Workspace** to perform various actions. You can also use the right-click menu to perform these actions.

Click through each icon below to view its description. These actions and the other right-click menu items are described in more detail later in this course.

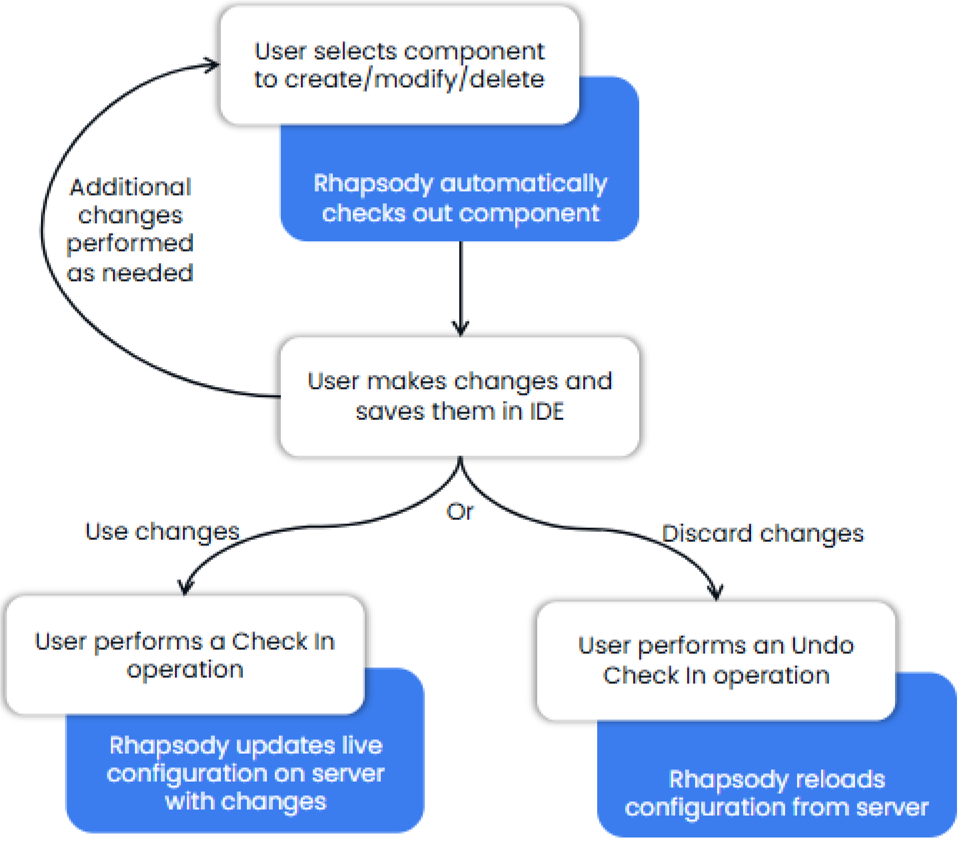
The Route Canvas

The Route Canvas provides a tabbed environment where you can view the layout of the currently selected route. Message flow through a route is always from the left of the canvas (the Input side) to the right (the Output side). Pathways are indicated by arrowed connectors between components. Whenever a pathway branches, a copy of the original message is created and sent along the second pathway.



Configuration Integrity

Rhapsody protects the integrity of a running system by controlling the way changes are made to the configuration. This is managed by ensuring that only one user can modify a component at a time using a check-out / check-in process. The general process for creating or modifying a component is:



Check out

The process of editing a component starts with a check-out.

- The check-out process prevents other users from making any changes to the component.
- The check-out process normally occurs automatically when the configuration for a component is changed in the IDE, or it can occur manually when you want to prevent changes occurring to a component.

Check in

The configuration held on the engine is used during message processing. Any changes made to the configuration of a component only take effect once that change has been checked in. When a check-in occurs:

- Processing is paused while the configuration is copied from the IDE to the engine.
- Once the configuration has been verified, processing resumes using the updated configuration.

- A message identifying the checked-in components and the success status of the action is displayed in the Information tab beneath the Route Canvas to confirm the check-in to the administrator.

If you attempt to close the IDE without checking in changes, the **Check In** dialog displays. You are requested to either check in the changes or to discard them before closing. This ensures that components of a configuration don't remain checked out when you are not actively working on them.

Undo check-out

At any point prior to initiating a check-in, you can reverse all changes made to checked-out components with the **Undo Check Out** option. Selecting this option copies the component's configuration from the engine back to the IDE, overwriting any changes made since check-out and restoring the component's configuration to the state it was in prior to check-out.

Rhapsody Management Console

he Rhapsody Management Console is a web-based monitoring tool that provides a view of the Rhapsody Engine's health and activity. You can view details such as:

- Number of messages processed by an individual route or communication point. Information from a component's log file identifies any problems or outages that may have occurred.
- Changes made to the structure and/or content of individual messages as they pass through a Rhapsody route. Specific properties (metadata) associated with a message can also be viewed with this tool.
- Statistical details relating to engine uptime and the total number of processed messages. This information can be used to show periods of high message throughput, thereby identifying potential bottlenecks which may slow message processing.
- Names and host identifiers of all administrators currently logged in to the Management Console. An option is available to disconnect logged-in users who are currently inactive.

The Management Console is supported on the latest version of all major browsers (Google Chrome, Microsoft Edge, Mozilla Firefox, & Safari).

To access the Management Console

You can access the Management Console in any of the following ways:

1. Select the Management Console icon in the IDE toolbar.

Rhapsody Management Console icon

2. Right-click the **Rhapsody Service Monitor** icon in the System Tray, then select **Open Management Console**.

System Monitor in running state icon

3. Open a browser and type the URL: https://<servername_name>. For example, https://localhost:8444.

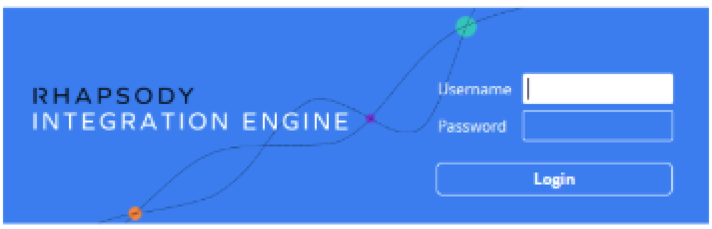
To login:

1. Enter a **Username** and **Password**.

   Your password is case-sensitive and is the same as the one you use to access the IDE.

2. Select **Login**.

Education

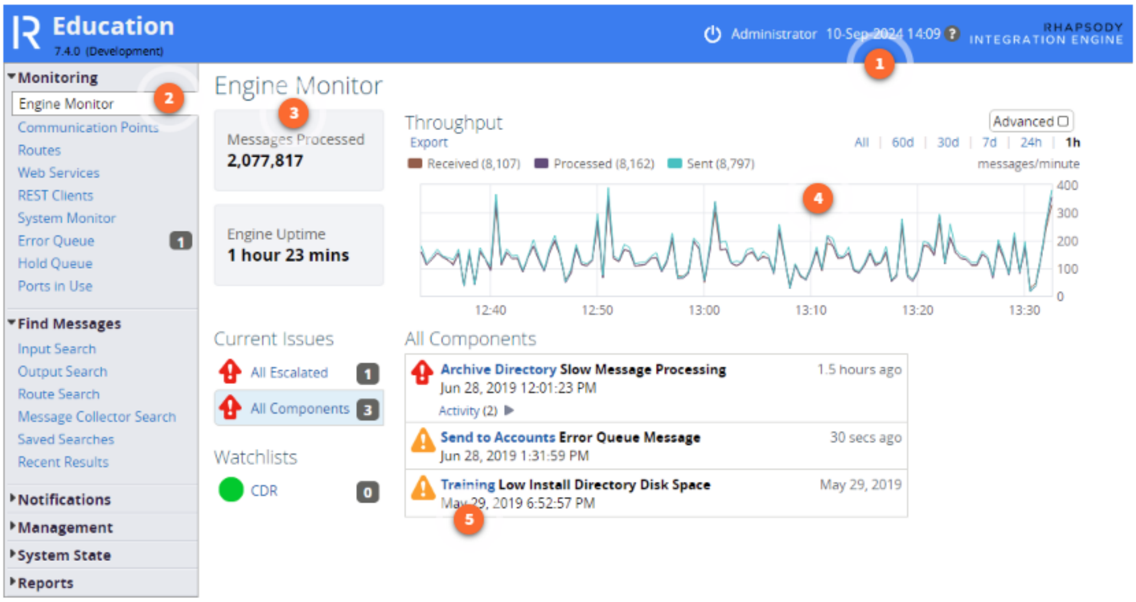


Management Console login

The Management Console

The initial page in the Management Console is the **Engine Monitor**. This page provides a snapshot of the number of hours since the engine was last started, the number of processed messages, a graphical display of engine throughput, and a list of the issues that might be affecting performance. The page automatically refreshes every 30 seconds.

The different areas of the Management Console are identified below:



Information in the Management Console

Select the headings on the Menu bar to access the following information:

- **Monitoring**
  View the messages processed by a specific communication point or route, or that are currently sitting in the **Error** or **Hold** queues.

- **Find Messages**
  Locate a message from the archive that was processed by a specific communication point or route. Save a set of useful search criteria for later re-use.

- **Notifications**
  Set up how an administrator is notified when specific events occur on the engine, such as a route or communication point shutting down in error.

- **Management**
  Manage the Rhapsody archive, message counters, lookup tables and other settings relating to the use of this component.

- **System State**
  View the full Rhapsody system and audit logs to learn the details of any error or other event that has occurred on the engine.

- **Reports**
  View reports detailing the speed at which messages are moving through the engine. This option is used to identify patterns in engine use or bottlenecks that are restricting message flow.

More Information

Additional information on this component along with several exercises can be found in "The Management Console" module later in this course.

Rhapsody Messaging Toolkit

The Rhapsody Messaging Toolkit is a set of desktop applications used to help define, parse, test, and transform messages in Rhapsody. They are standalone tools that can work independently of Rhapsody. In fact, many customers have licensed only these tools for use on their site. They are, however, specifically adapted to support a Rhapsody solution, especially when the creation and modification of HL7 messages is a requirement.

The HL7 V2 standard, for example, has a well-defined hierarchical structure which allows fields and the content they contain to be reliably located within a message and for this information to be used to control further processing. HL7 message are discussed in the "Working with Messages" module.

Components

The Rhapsody Messaging Toolkit includes:

- EDI Message Designer
  Define and modify the layout of a structured message.

- EDI Explorer
  Create and validate test messages based on a definition file created using the EDI Message Designer.

- Map Designer
  Transform the structure of a message from one format to another.

We will go into much more detail about each component later in this course.

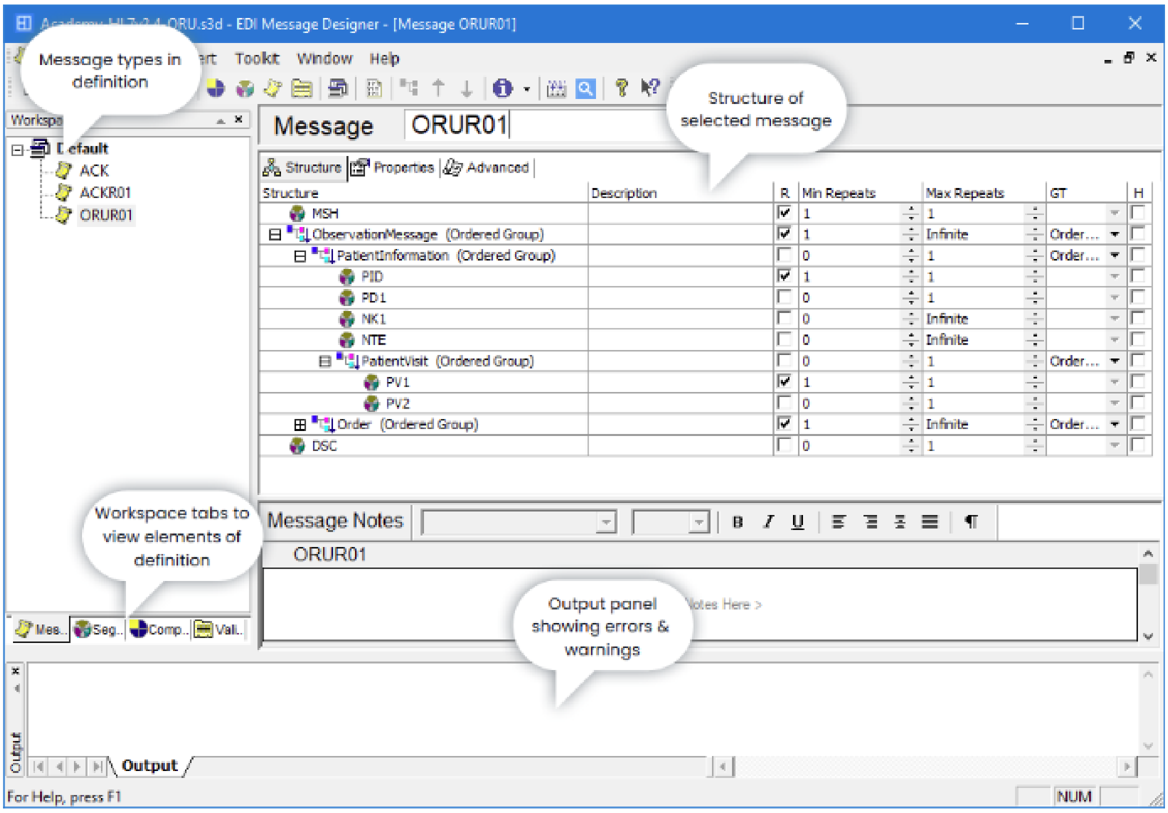
EDI Message Designer

The EDI Message Designer application is used to create and manage message definitions having an .s3d filename extension. These definitions hold information describing the structure (but not the content) of a message. These are typically used to define HL7 messages. This information is used by Rhapsody to parse messages during processing. This module focuses on the use of this application with HL7 messages, which are an industry standard for messages containing healthcare-related information.

To access the EDI Message Designer

The EDI Message Designer can be launched from either the Rhapsody IDE or the Windows Start menu (search for “EDI Message Designer”). No logon is required.

The window below is loaded with an HL7 ORUR01 (Observation) message definition:



EDI message designer

Workspace tabs

The **Workspace** tabs list the components contained in the currently loaded message definition, and include:

- **Message**
  The messages contained in the selected definition; such as ACK (Acknowledgement), ADT (Admit, Discharge and Transfer) and ORM (General Order) messages.
- **Segments**
  The segments contained in a selected message, such as MSH (Message Header), EVN (Event) and PID (Patient Identification) segments.

- **Composites**
  The complex fields contained within a segment, such as MSG (Message Type), XPN (Patient Name) and TS (Date Time).

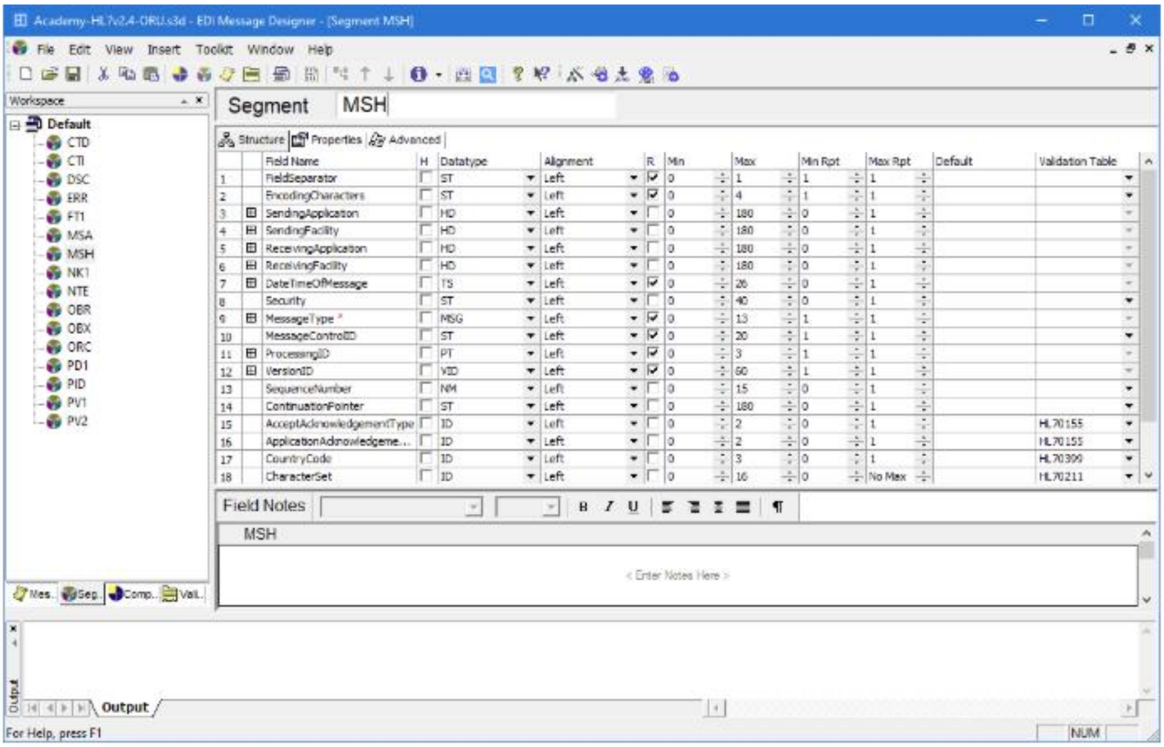
- **Validation Tables**
  The validation tables included with a message used to validate its data, such as HL70001 (Patient Gender), HL70078 (Abnormality Flags) and HL70399 (Country Codes).

The Output panel

The **Output** panel lists messages from the application, which may include alerts and warnings, as specific actions are taken, such as adding a new message type to the current definition.

The Details panel

The **Details** panel lists the structure of the selected message component. The screenshot below, for example, shows the fields contained within the structure of the MSH (Message Header) segment in an HL7 ORU message. The **Field Notes** panel is included in the screenshot, though this is normally hidden by clearing the selection of the **Notes** option in the **View** menu.



EDI message designer – MSH details

The Library Wizard

The EDI Message Designer ships with a library of pre-defined message definitions. These include definitions for several message types including HL7 V2. The libraries can be used to help build definitions for use in Rhapsody.

More Information

Using the EDI Message Designer to create and modify HL7 message definitions is described in more detail in the "Working with Messages" and "Interpreting Message Processing" modules.

EDI Explorer

EDI Explorer is used to create and validate test messages based on a message definition created by EDI Message Designer. The test messages are then used to confirm that a definition can parse the messages correctly and that the engine processes messages in the expected way.

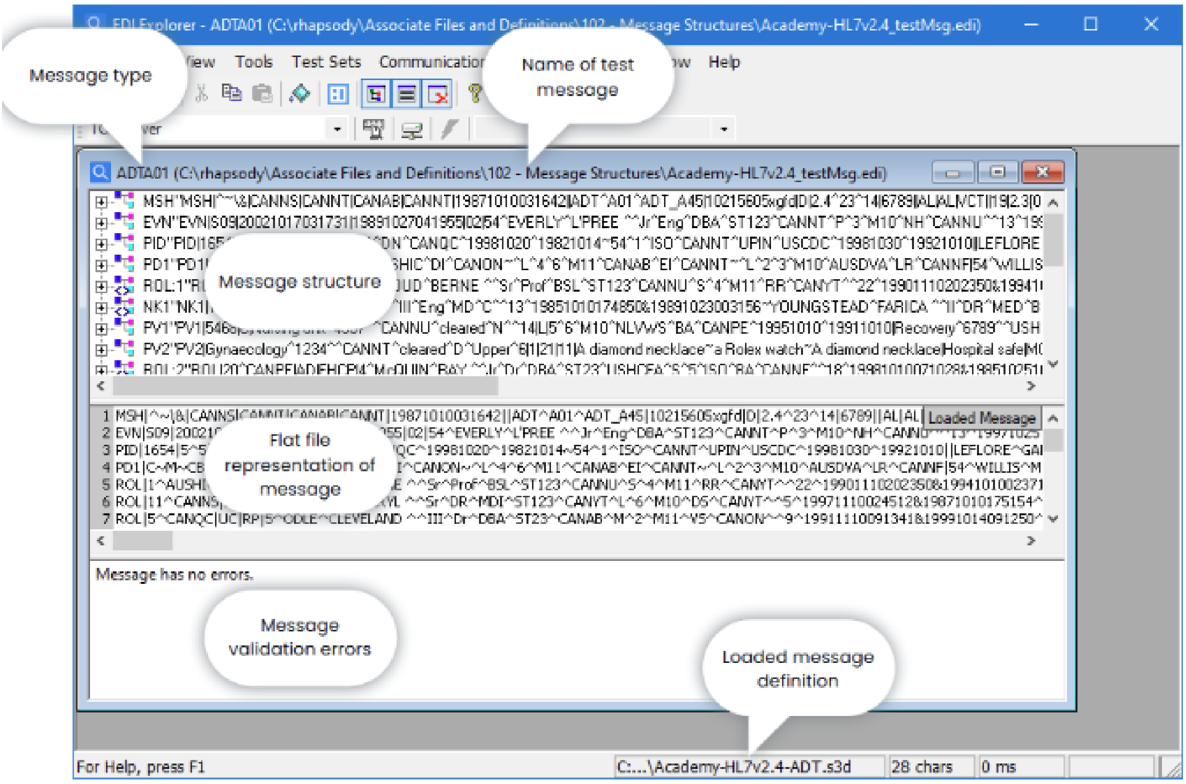
To access the EDI Explorer

The EDI Explorer can be launched from either the Rhapsody IDE, the EDI Message Designer (when launched this way, the current message definition is automatically loaded) or from the Windows Start menu (search for "EDI Explorer"). No logon is required.

Using EDI Explorer

Before using the application to create or validate a test message, a previously created message definition must be loaded. To load a message definition, select **Open Definition** from the **File** menu. Once loaded, the definition's filename appears in the status bar at the bottom of the screen.

A test message can be imported or created from scratch. The following screenshot shows an ADTA01 (Admit, Discharge, Transfer) message based on the HL7 V2.4 standard:



EDI explorer

The three parts of a message's Message Panel:

- Message structure
  Displays the message's structure along with its content. Expand the individual segments to show their fields and composites. Select the individual fields within a message to update their content.

- Flat file message
  Displays the message as it will be transmitted, and as it would appear if opened in a text editor (such as Notepad). The flat file representation of a message can be exported from the File menu.

- Errors and Warnings
  Data in a test message is checked for a maximum and minimum length and, in some cases, for a match with values in a **Validation Table**. Any discrepancies will be reported in the Errors and Warnings panel.

Connecting Via TCP

EDI Explorer can be configured to mimic an external system set up as either a TCP Server or TCP Client. It can then be used to send test messages into Rhapsody and, if applicable, receive the appropriate responses.

More Information

We will use EDI Explorer to create and modify test messages, then send them into a route for processing, in many of the upcoming modules in the course.

Map Designer

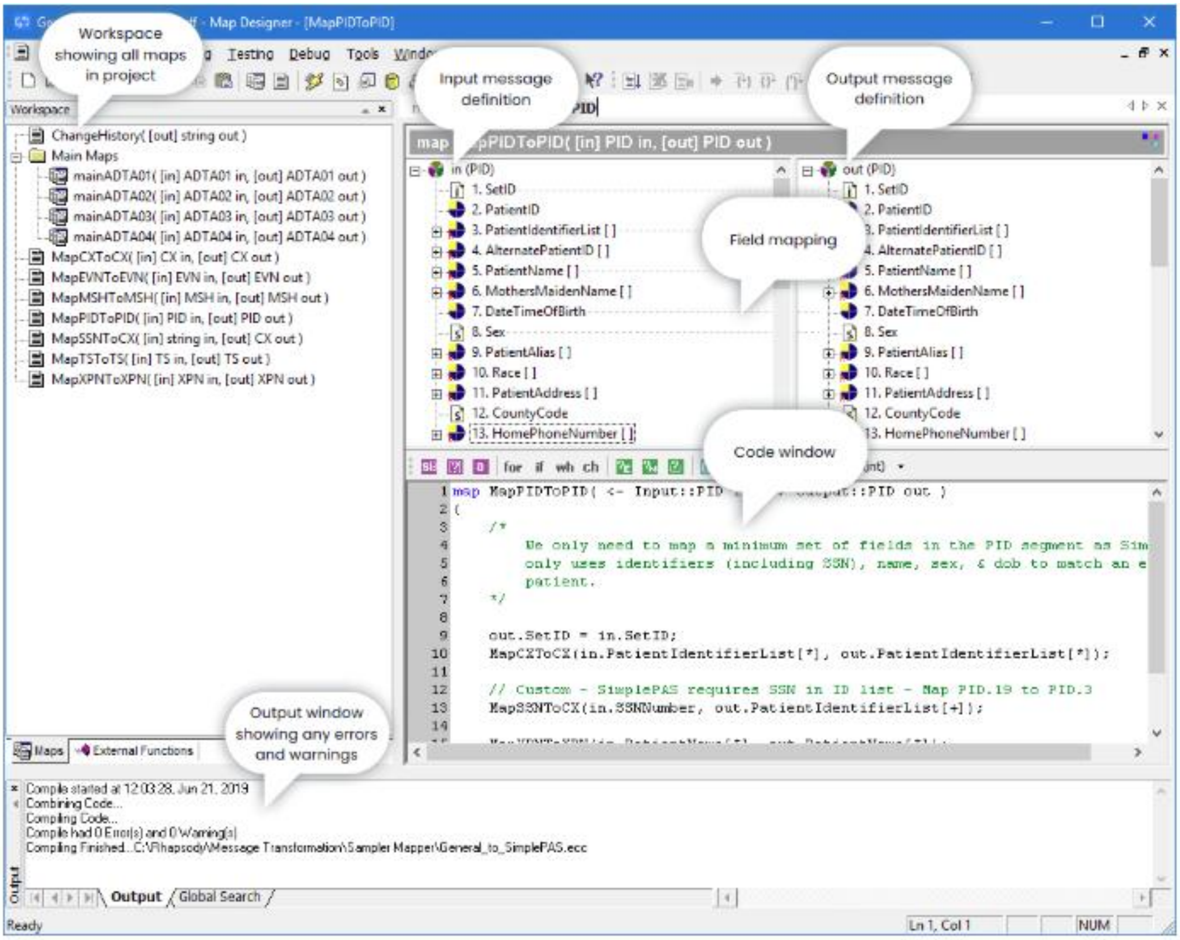
The Map Designer supports the creation of rules and transformations that control how information in an incoming message is mapped to corresponding fields in an outgoing message. It allows mapping between messages of the same type, for example HL7 V2.3 messages to HL7 V2.4, or of different types, for example messages in CSV format to messages in the FHIR JSON format.

To access the Map Designer

The Map Designer can be launched from either the Rhapsody IDE or from the Windows Start menu (search for “Map Designer”). No logon is required.

The Mapper screen

A new mapper project requires the provision of input and output message definitions, typically but not necessarily created using EDI Message Designer. Creating the actual mapping involves dragging a message field from an input message and dropping it to its corresponding location in the output message. As the connection is made, mapping code is automatically generated in the Code window, where it can be amended and documented as required.



Map designer

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HL70001 (Patient Gender), HL70078 (Abnormality Flags) and HL70399 (Country Codes).

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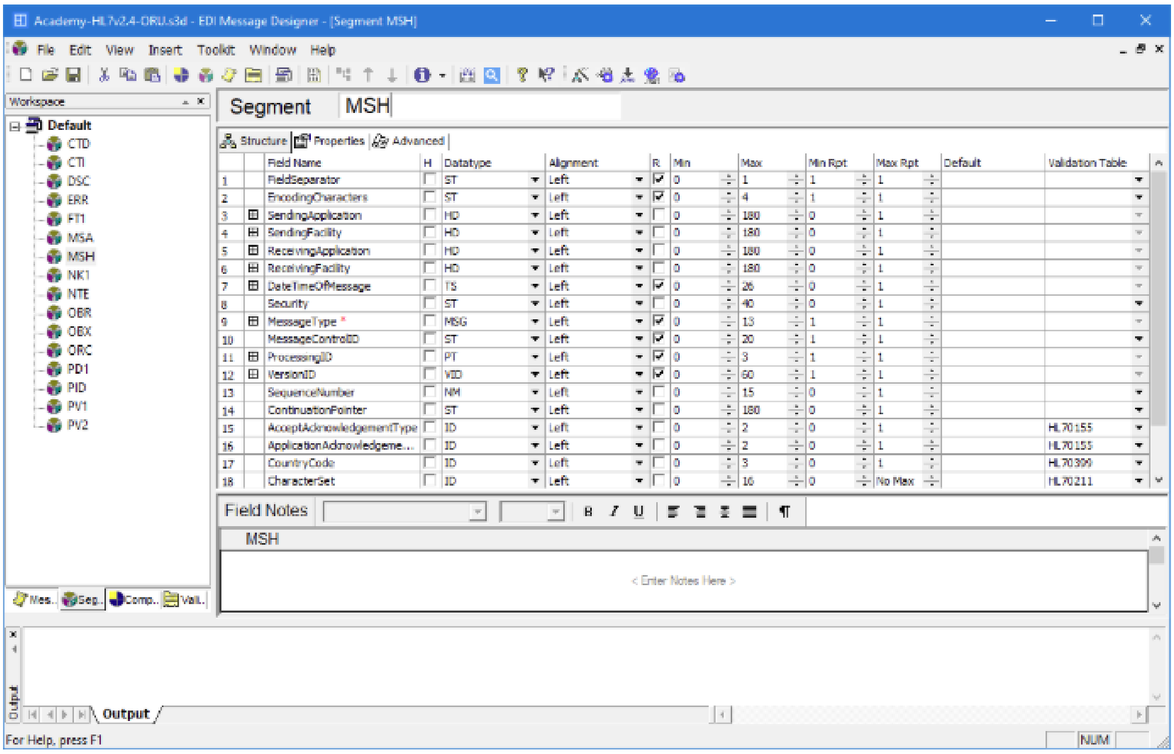

EDI message designer - MSH details

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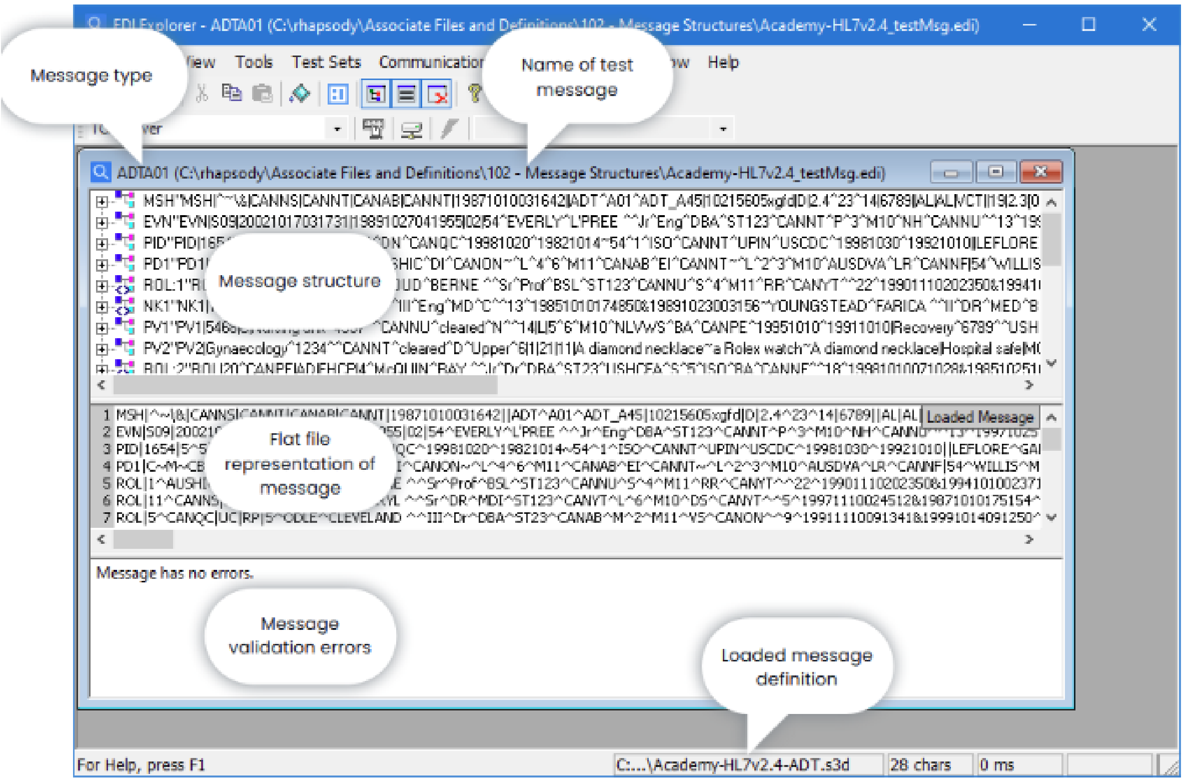
Using EDI Explorer

Before using the application to create or validate a test message, a previously created message definition must be loaded. To load a message definition, select **Open Definition** from the **File** menu. Once loaded, the definition's filename appears in the status bar at the bottom of the screen.

A test message can be imported or created from scratch. The following screenshot shows an ADTA01 (Admit, Discharge, Transfer) message based on the HL7 V2.4 standard:



EDI explorer

The three parts of a message's Message Panel:

- Message structure
  Displays the message's structure along with its content. Expand the individual segments to show their fields and composites. Select the individual fields within a message to update their content.
- Flat file message
  Displays the message as it will be transmitted, and as it would appear if opened in a text editor (such as Notepad). The flat file representation of a message can be exported from the File menu.
- Errors and Warnings
  Data in a test message is checked for a maximum and minimum length and, in some cases, for a match with values in a **Validation Table**. Any discrepancies will be reported in the Errors and Warnings panel.

Connecting Via TCP

EDI Explorer can be configured to mimic an external system set up as either a TCP Server or TCP Client. It can then be used to send test messages into Rhapsody and, if applicable, receive the appropriate responses.

More Information

We will use EDI Explorer to create and modify test messages, then send them into a route for processing, in many of the upcoming modules in the course.

Map Designer

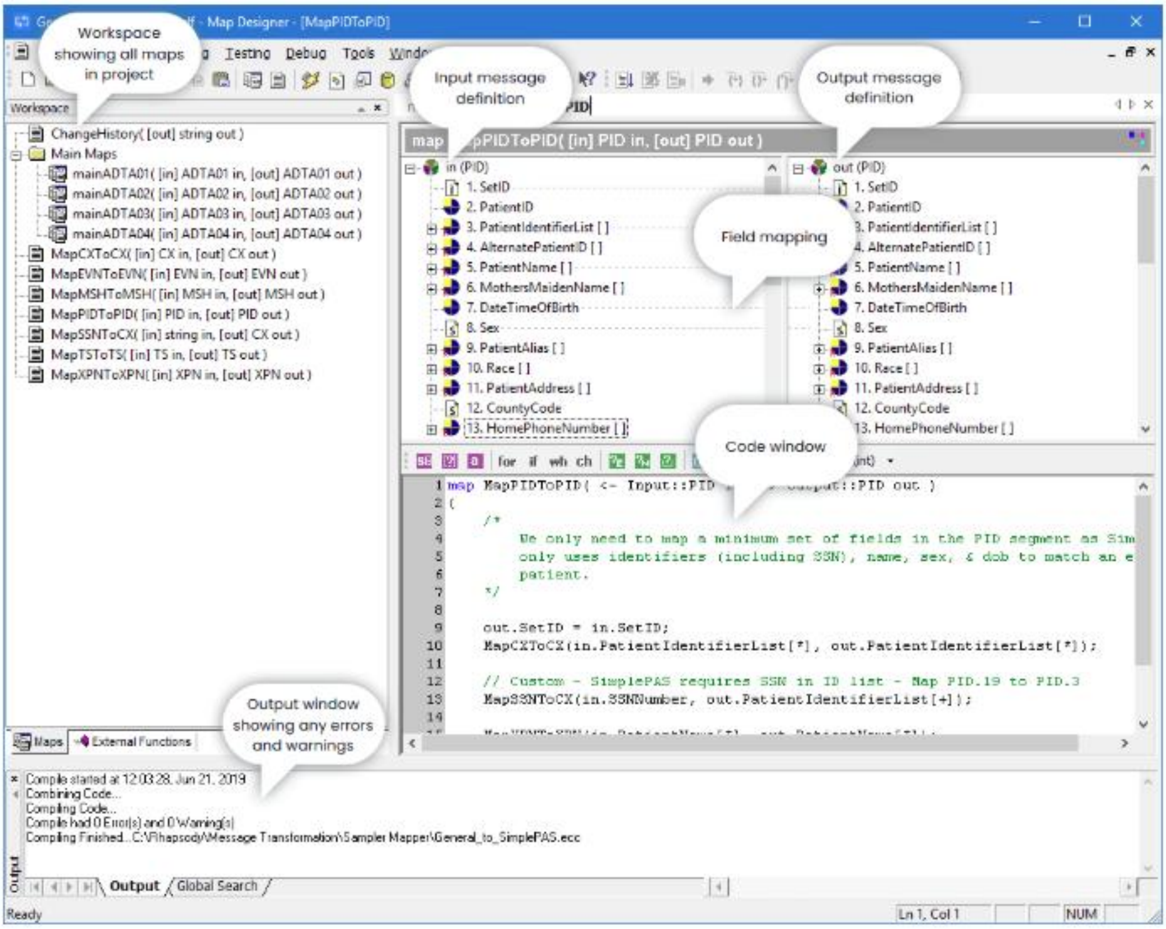
The Map Designer supports the creation of rules and transformations that control how information in an incoming message is mapped to corresponding fields in an outgoing message. It allows mapping between messages of the same type, for example HL7 V2.3 messages to HL7 V2.4, or of different types, for example messages in CSV format to messages in the FHIR JSON format.

To access the Map Designer

The Map Designer can be launched from either the Rhapsody IDE or from the Windows Start menu (search for “Map Designer”). No logon is required.

The Mapper screen

A new mapper project requires the provision of input and output message definitions, typically but not necessarily created using EDI Message Designer. Creating the actual mapping involves dragging a message field from an input message and dropping it to its corresponding location in the output message. As the connection is made, mapping code is automatically generated in the Code window, where it can be amended and documented as required.



Map designer

More Information

A tested and saved Mapper definition (saved as an .mdf file) is loaded into the engine, checked in, then associated with a Mapper filter before it can become a part of a Rhapsody solution. How this is done, along with more detail on creating a mapper definition, is described in the "Message Transformation" module later in this course.